

Riccardo Majellaro

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EXPERIENCE

MACHINE LEARNING ENGINEER | DUCKDUCKGOOSE

Jun 2024 – Present | Delft, Netherlands

- Research on deepfake detection and explainability methods.
- Designed and conducted experiments to improve internal and academic SOTA deepfake detection models, achieving up to 15% accuracy improvement on fake samples at very low false positive rate (~0.5%) while reducing model size 6.5x and inference speed 2.5x.
- Led research and development of novel explainability method for deepfake detection models, outperforming current academic SOTA.
- Wrote production-level code to deploy ML models, optimizing speed and memory requirements.
- Designed algorithm to minimize models deployment costs while maintaining efficiency by leveraging clients usage statistics.

TEACHING ASSISTANT | LEIDEN UNIVERSITY

Feb 2023 - Jul 2023 | Leiden, Netherlands

- MSc course “Reinforcement Learning” taught by Prof. Aske Plaat.
- Correcting and grading assignments. Helping students via email and during workgroup sessions.

PUBLICATIONS

EXPLICITLY DISENTANGLED REPRESENTATIONS IN

OBJECT-CENTRIC LEARNING | PUBLISHED AT TMLR | 2024

[Riccardo Majellaro](#), [Jonathan Collu](#), [Aske Plaat](#), [Thomas M. Moerland](#)

SLOT STRUCTURED WORLD MODELS | PREPRINT | 2024

[Jonathan Collu](#), [Riccardo Majellaro](#), [Aske Plaat](#), [Thomas M. Moerland](#)

PROJECTS

INTERACTION INFORMATION OPTIMIZATION FOR

OBJECT-CENTRIC REPRESENTATION LEARNING | REPORT

- Generalized the objective presented in “Information-Theoretic Segmentation by Inpainting Error Maximization” from 1 to N foreground objects and applied it to Slot Attention.

TRASHAWAY ROBOT | REPO

- Trained with deep reinforcement learning a robot to perform the task of “cleaning” an ad hoc simulated environment from cubes, using a camera as its only sensor.
- Successfully deployed the trained agent in the real world (using a PiCar-X), completing 100% of the task on the first attempt.

TEXT-TO-IMAGE ADDITIONS/SUBTRACTIONS SOLVER | REPO

- Trained a Recurrent Neural Network (RNN) to solve additions and subtractions for numbers up to three digits.

ADVERSARIAL ATTACKS ON VISION MODELS | REPO

- Developed an open-source library to generate adversarial attacks through gradient-based methods and evolutionary strategies, enabling robustness benchmarking for neural networks.

EDUCATION

MSC COMPUTER SCIENCE

Artificial Intelligence track

LEIDEN UNIVERSITY

Sep 2021 - Jul 2023

- GPA: 8.9/10 Cum Laude (with honors)
- Top-graded thesis in CS department

BSC COMPUTER ENGINEERING

UNIVERSITY OF MODENA AND

REGGIO EMILIA, ITALY

Sep 2017 - Oct 2020

- GPA: 8.8/10
- Thesis: “Distributed Training of DETR on Marconi100” - Scaled and analyzed the parallel training of the Detection Transformer by Meta on MARCONI100, ranked 9th world's most powerful supercomputer in 2020.

SKILLS

PROGRAMMING

Main expertise: Python • C • SQL

Past experience: Java • Javascript • C++

FRAMEWORKS / TOOLS

PyTorch • TensorFlow • PyTorch3D • NumPy • Scikit-learn • Matplotlib • OpenCV • Pandas • Git • Unix shell • Slurm • LaTeX

LANGUAGES

Italian (native) • English (fluent)

AI / ML CONCEPTS

Computer Vision • Natural Language Processing (NLP) • Reinforcement Learning • Unsupervised Learning • Self-supervised Learning • Optimization • Explainability (XAI) • Dimensionality Reduction • Adversarial Attacks • Generative AI • LLMs • MLOps

RESEARCH INTERESTS

My primary research interests are in machine learning and computer vision, with a focus on representation learning, unsupervised learning, and self-supervised learning. I worked on topics such as object-centric representation learning, multi-object 3D scene generation, and (model-based) reinforcement learning. I am also interested in vision-language models, embodied agents, and external memories for artificial neural networks. Currently working on deepfake detection.